Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Hons.) Final Examination -2022
Answer Script: Electronic Devices and Circuit-II (CSE-207)

Question 1 (Short Questions)

a) State the meaning of virtual ground.

The virtual ground concept applies to an ideal Operational Amplifier (Op-Amp) operating in the closed-loop (negative feedback) configuration. It states that if the non-inverting input (+) is connected to ground (0V), the high open-loop gain forces the inverting input (-) to also be approximately 0V. The two inputs are not physically connected, hence "virtual."

b) Mention the working principle of voltage buffer.

A voltage buffer (or voltage follower) uses a non-inverting Op-Amp configuration with 100% negative feedback (output connected directly to the inverting input). It provides a voltage gain of 1 (Unity Gain), very high input impedance, and very low output impedance. Its purpose is to act as an impedance transformer, isolating a high-impedance source from a low-impedance load without attenuating the signal.

c) How can we find the cutoff frequency?

The cutoff (or corner) frequency is the frequency at which the output power drops to half its mid-band value (i.e., -3 dB). It is found by analyzing the RC networks in the circuit. For a standard low-pass or high-pass RC filter, it is calculated using $f_c = \frac{1}{2\pi RC}$. For an Op-Amp, it is where the gain drops by 3 dB from the passband.

d) List the basic features of Op-Amp.

1. Very High Voltage Gain (Open-loop gain $A_{OL} \approx \infty$).
2. Very High Input Impedance ($Z_{in} \approx \infty$).
3. Very Low Output Impedance ($Z_{out} \approx 0$).
4. High Common Mode Rejection Ratio (CMRR).

5. Large Bandwidth (ideally infinite).

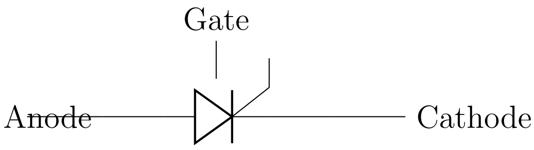
e) Mention the applications of photodiode.

1. Light meters (photography).
2. Optical communication receivers (fiber optics).
3. Smoke detectors.
4. Solar cells (power generation).
5. Medical imaging equipment (CT scanners).

f) Sketch the terminal identification of SCR.

A Silicon Controlled Rectifier (SCR) has three terminals:

- **Anode (A):** Positive terminal.
- **Cathode (K):** Negative terminal.
- **Gate (G):** Control terminal.



Question 2 (Three out of Four)

a) i. Differentiate between Hartley and Colpitts oscillator.

Feature	Hartley Oscillator	Colpitts Oscillator
Tank Circuit	Uses two inductors (L1, L2) and one capacitor (C).	Uses two capacitors (C1, C2) and one inductor (L).
Feedback	Voltage divider action of inductors (tapped coil).	Voltage divider action of capacitors (capacitive divider).
Frequency	$f_r = \frac{1}{2\pi\sqrt{(L1+L2)C}}$	$f_r = \frac{1}{2\pi\sqrt{L\left(\frac{C1C2}{C1+C2}\right)}}$

ii. Which type of material is used for infrared LED?

Gallium Arsenide (GaAs). Unlike standard LEDs (Gallium Phosphide) which emit visible light, GaAs has a bandgap energy that releases energy in the infrared spectrum (invisible to the human eye).

b) Formalize the characteristics and applications of Schottky Diode.

Characteristics:

1. **Low Forward Voltage Drop:** Typically 0.2V - 0.3V (compared to 0.7V for Si).
2. **Very Fast Switching Speed:** No minority carrier storage (it is a majority carrier device), so zero reverse recovery time.

3. **High Reverse Leakage Current:** Higher than standard PN diodes.

Applications:

1. High-frequency rectifiers (Switching power supplies, RF detectors).
2. Clamping diodes (to prevent transistor saturation).
3. Solar cell bypass diodes.

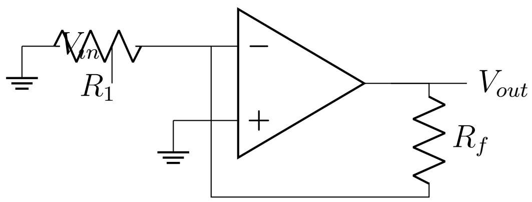
c) Discuss the function of using an inverting constant-gain multiplier. Draw the diagram.

Function: An inverting constant-gain multiplier is an Op-Amp configuration designed to amplify an input signal by a precise factor (Gain) while shifting its phase by 180 degrees. The gain is set solely by the ratio of two resistors (R_f and R_1), making it very stable and independent of temperature variations.

Formula:

$$V_{out} = -\left(\frac{R_f}{R_1}\right) V_{in}$$

Diagram:



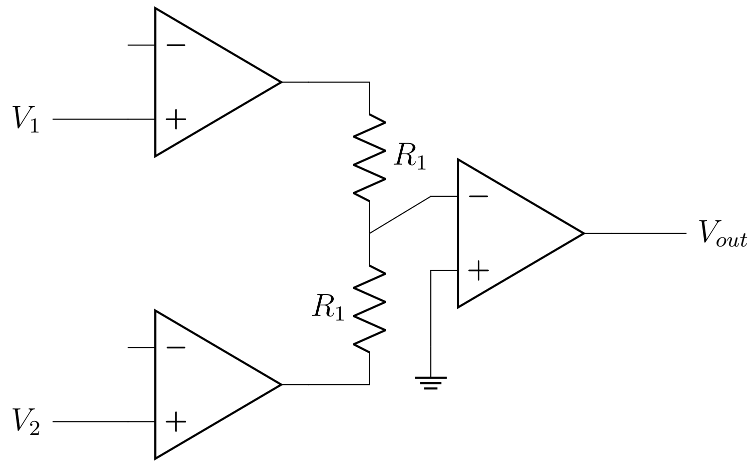
d) Explain the basic Instrumentation amplifier model with proper figure.

Explanation: An instrumentation amplifier is a differential voltage amplifier (subtractor) designed for high accuracy. It features very high input impedance (unlike a simple Op-Amp differential amplifier), high CMRR, and adjustable gain.

Model: It consists of 3 Op-Amps.

1. **Input Stage:** Two non-inverting Op-Amps (Buffers). They provide extremely high input impedance and differential gain.
2. **Output Stage:** A standard differential amplifier (Subtractor) that rejects common-mode signals and converts the differential signal to a single-ended output.

Diagram:



Question 3 (Three out of Four)

a) Determine the voltage gain, input, and output impedance with feedback.

Given: $A = -50$, $R_i = 5k\Omega$, $R_o = 10k\Omega$, $B = -0.10$.
Feedback Type: Voltage-series (Series-Shunt) negative feedback.

Calculations:

1. **Voltage Gain with feedback (A_f):**

$$A_f = \frac{A}{1 + AB}$$
$$1 + AB = 1 + (-50)(-0.10) = 1 + 5 = 6$$
$$A_f = \frac{-50}{6} \approx -8.33$$

2. **Input Impedance with feedback (Z_{if}):**

$$Z_{if} = R_i(1 + AB) = 5k\Omega \times 6 = 30k\Omega$$

3. **Output Impedance with feedback (Z_{of}):**

$$Z_{of} = \frac{R_o}{1 + AB} = \frac{10k\Omega}{6} \approx 1.67k\Omega$$

b) Calculate the cutoff frequency of an op-amp.

Given: $B_1 = 2$ MHz (Unity Gain Bandwidth), $A_{VD} = 200$ V/mV.

$$A_{VD} = 200 \text{ V/mV} = 200 \times 10^3 = 200,000$$
$$f_{unity} = B_1 = 2 \text{ MHz} = 2 \times 10^6 \text{ Hz}$$

Formula: Gain-Bandwidth Product is constant.

$$A_{VD} \times f_c = f_{unity}$$

Calculation:

$$f_c = \frac{f_{unity}}{A_{VD}} = \frac{2 \times 10^6}{200,000} = 10 \text{ Hz}$$

c) Determine the quiescent levels of I_{CQ} and V_{CEQ}.

Exact Analysis (Thevenin Equivalent):

- Thevenin Voltage: $V_{Th} = V_{CC} \times \frac{R_2}{R_1+R_2}$
- Thevenin Resistance: $R_{Th} = R_1 \parallel R_2$
- Base Current: $I_{BQ} = \frac{V_{Th}-V_{BE}}{R_{Th}+(\beta+1)R_E}$
- Collector Current: $I_{CQ} = \beta I_{BQ}$
- Collector-Emitter Voltage: $V_{CEQ} = V_{CC} - I_{CQ}(R_C + R_E)$

Approximate Analysis ($\beta R_E \geq 10R_2$):

- Base Voltage: $V_B \approx V_{CC} \times \frac{R_2}{R_1+R_2}$
- Emitter Voltage: $V_E = V_B - V_{BE}$
- Emitter Current: $I_E = \frac{V_E}{R_E}$
- Collector Current: $I_{CQ} \approx I_E$
- Collector-Emitter Voltage: $V_{CEQ} = V_{CC} - I_{CQ}(R_C + R_E)$

d) UJT Questions.

i. Does UJT work on AC or DC?
Answer: DC. A Unijunction Transistor (UJT) requires a DC voltage supply (V_{BB}) between the two bases to establish the interbase resistance. It is a DC-triggered device used primarily in DC oscillator circuits (Relaxation oscillators).

ii. Which type of material is the channel of a unijunction transistor made up of?
Answer: IV. N type.
A UJT consists of a bar of **N-type** silicon (the channel). A P-type alloy is fused to the side to form the Emitter junction.

Question 4 (Any Three)

4. a) i. How to make a Darlington pair transistor?

Answer: Connect two BJTs (either both NPN or both PNP) such that the emitter of the first transistor drives the base of the second transistor.

- **Connection:** Collector of Q1 connects to Collector of Q2. Emitter of Q1 connects to Base of Q2.
- **Result:** The pair acts as a single transistor with a current gain $\beta_D \approx \beta_1 \times \beta_2$ and a high input impedance.

ii. For the Voltage feedback network, determine I_C, V_C, V_{CE}.

For a Collector Feedback Bias circuit:

- Base Current: $I_B = \frac{V_{CC}-V_{BE}}{R_B+R_C}$

- Collector Current: $I_C = \beta I_B$
- Collector Voltage: $V_C = V_{CC} - I_C R_C$
- Collector-Emitter Voltage: $V_{CE} = V_C$ (Since Emitter is grounded)

4. b) Wien Bridge Oscillator Frequency.

Given: $R1 = 2k\Omega$, $R2 = 4k\Omega$, $C1 = 1\mu F$, $C2 = 2\mu F$.

Formula:

$$f_o = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$$

Calculation:

$$f_o = \frac{1}{2\pi\sqrt{(2\times10^3)(4\times10^3)(1\times10^{-6})(2\times10^{-6})}}$$

$$f_o = \frac{1}{2\pi\sqrt{16\times10^{-6}}} = \frac{1}{2\pi(4\times10^{-3})} = \frac{1}{0.02513} \approx 39.79 \text{ Hz}$$

4. c) Calculate the output voltage for the circuit.

Given: $V_1 = 50$ mV sin(1000 t), $V_2 = 10$ mV sin(3000 t).

Formula (Inverting Summing Amp):

$$V_{out} = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2\right)$$

Calculation (Assuming $R_f = R_1 = R_2$):

$$V_{out} = -(V_1 + V_2) = -[50\sin(1000t) + 10\sin(3000t)] \text{ mV}$$

4. d) i. Analyze the comparison between SCR and GTO.

Feature	**SCR**	**GTO**
Turn-off	Cannot be turned off via Gate. Requires commutation.	Can be turned off by applying a negative pulse to the Gate.
Control	Latching switch (DC control of AC).	Full control (ON and OFF via Gate).
Switching Speed	Slower turn-off.	Faster turn-off capability.
Complexity	Simpler drive circuit (positive pulse only).	More complex drive circuit (requires both positive and negative pulses).

ii. Draw the schematic symbol of DIAC.

Question 5 (Any Two)

5. a) Determine currents and voltages for common-base configuration.

Given: $V_{EE}, V_{CC}, R_E, R_C, \beta, V_{BE}$.

Calculations:

1. Emitter Current: $I_E = \frac{V_{EE} - V_{BE}}{R_E}$
2. Collector Current: $I_C = \alpha I_E \approx I_E$
3. Base Current: $I_B = \frac{I_C}{\beta}$
4. Collector Voltage: $V_C = V_{CC} - I_C R_C$
5. $V_{CE} = V_C - V_B$ (Since Base is grounded, $V_B = 0$, so $V_{CE} = V_C$)
6. $V_{CB} = V_C - V_B = V_C$

5. b) i. Differentiate between class-C and class-D amplifier.

Feature	**Class-C Amplifier**	**Class-D Amplifier**
Conduction Angle	Less than 180° (usually 90°).	Switching mode (ON or OFF).
Operation	Linear (but highly distorted).	Non-linear (Pulse Width Modulation).
Efficiency	High (approx. 80-90% theoretically).	Very High (approx. 90-100% theoretically).
Applications	RF oscillators and tuned amplifiers.	Battery-powered devices, Hi-Fi audio, subwoofers.

ii. Which type of filter circuit given here?

Answer (General): If the circuit has an Op-Amp and RC components, it is an **Active Filter**.

- **Low-Pass Filter (LPF):** Passes low frequencies and blocks high frequencies (capacitor in feedback path).
- **High-Pass Filter (HPF):** Passes high frequencies and blocks low frequencies (capacitor at input).

Working Procedure: The filter uses the Op-Amp's high input impedance and gain to selectively amplify or attenuate specific frequency bands determined by the reactance of the capacitors and the values of the resistors.

5. c) i. Specify the input offset voltage.

Answer: Input Offset Voltage (V_{IO}) is the differential DC voltage required between the input terminals of an Op-Amp to make the output voltage exactly 0V (when it is supposed to be 0V). In real Op-Amps, mismatches inside the transistor pairs cause a small internal voltage (typically 1-5 mV for bipolar, 1-20 mV for FET) that acts like a small battery in series with one input, causing output error even with inputs shorted.

ii. Calculate Input power, Output power, and Efficiency.

Given: $R_L = 1k\Omega$, $R_C = 2.5k\Omega$, $V_{CC} = 15V$, $V_{BE} = 0.7V$, $V_{CE} = 5V$. Base current peak = 20 mA.

Calculations:

1. **DC Input Power:**

$$I_C = \frac{V_{CC} - V_{CE}}{R_C} = \frac{15 - 5}{2.5k\Omega} = 4 \text{ mA}$$

$$P_{in(DC)} = V_{CC} \times I_C = 15V \times 4mA = 60 \text{ mW}$$

2. **AC Output Power:**

$$V_{out(peak)} = I_{C(peak)} \times R_L = 4mA \times 1k\Omega = 4V$$

$$V_{out(rms)} = \frac{V_{peak}}{\sqrt{2}} = \frac{4}{1.414} \approx 2.83V$$

$$P_{out(AC)} = \frac{V_{out(rms)}^2}{R_L} = \frac{(2.83)^2}{1000} \approx 8 \text{ mW}$$

3. **Efficiency:**

$$\eta = \frac{P_{out(AC)}}{P_{in(DC)}} \times 100\% = \frac{8mW}{60mW} \times 100\% \approx 13.33\%$$